

Dynamics of the QCD Vacuum Condensate Amplitude in Dense Matter and Cosmology

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Abstract

Within the theoretical frameworks of Null-Vector Gravity (NVG) and the Vacuum Mass Fraction (VMF), the vacuum condensate is described by a complex scalar field $\Phi(x)$. Using a polar representation, the field is parameterized by a real radial mode $\mathcal{W}(x)$, which governs the in-medium effective mass of hadrons (M^*), and an angular mode $\theta(x)$ representing the Goldstone phase. This work provides a rigorous mathematical derivation of the classical equations of motion for the real amplitude field $\mathcal{W}(x)$ and its gauge-invariant coupling to the baryon current and vector meson fields in a dense nuclear medium. The coupling parameters are calibrated via QCD pion-nucleon ($\sigma_{\pi N} \approx 44$ MeV) and strange ($\sigma_{sN} \approx 30$ MeV) sigma-terms. The energy-momentum tensor is derived, and its Friedmann-Lemaître-Robertson-Walker (FLRW) reduction is performed. We demonstrate that the melting of the vacuum condensate ($\mathcal{W} \rightarrow 0$) under extreme compression leads to a violation of the Strong Energy Condition (SEC). This generates a quadratic density correction to the energy density, yielding a modified Friedmann equation that triggers a smooth cosmological bounce and eliminates the Big Bang singularity. Finally, the limits of the mean-field approximation are discussed using the Ginzburg criterion, local spatial fluctuations, and Coleman-Weinberg radiative corrections.

Keywords: vacuum condensate, Null-Vector Gravity, scalar field dynamics, cosmological bounce, Strong Energy Condition violation, QCD sigma-terms

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1 Introduction

The problem of mass generation in Quantum Chromodynamics (QCD) shows that the physical masses of ordinary hadrons, such as nucleons, are not simply the sum of the current masses of their constituent valence quarks. Instead, approximately 90% of the hadron mass is generated nonperturbatively by gluon field energy, confinement, and the trace anomaly of the energy-momentum tensor [1, 2], which is phenomenologically realized in low-energy models such as the MIT bag model [4]. In the vacuum, this nonperturbative component is intimately related to the chiral and gluon vacuum expectation values (VEVs).

Within the framework of Null-Vector Gravity (NVG) and the Vacuum Mass Fraction (VMF) model [3], the vacuum condensate is modeled macroscopically by a complex scalar field $\Phi(x)$. In a polar representation, this field is decomposed as:

$$\Phi(x) = \frac{1}{\sqrt{2}} \mathcal{W}(x) e^{i\theta(x)} \quad (1)$$

where $\mathcal{W}(x) \geq 0$ represents the real radial mode (amplitude of the vacuum condensate) and $\theta(x)$ is the real angular mode (Goldstone phase). The gradient of the phase $\partial_\mu \theta$ defines a preferred time direction (collinear with the 4-velocity of the nuclear medium), while the amplitude $\mathcal{W}(x)$ determines the effective masses of hadrons in the medium. Specifically, the in-medium effective mass of a hadron (such as a nucleon or the Ω -baryon) scales directly with the local condensate amplitude:

$$M^*(n_B) = g_s \mathcal{W}(n_B) \quad (2)$$

where g_s is the effective scalar coupling constant. In the vacuum (baryon density $n_B = 0$), the hadron mass is $M_0 = g_s \mathcal{W}_0$. Thus, the melting of hadron masses in a dense medium is governed by the ratio of the condensate amplitudes:

$$\frac{M^*(n_B)}{M_0} = \frac{\mathcal{W}(n_B)}{\mathcal{W}_0} \quad (3)$$

This paper is structured as follows. In Section 2, we present the action of the scalar sector and discuss its gauge invariance. Section 4 derives the classical equations of motion. In Section 5, we formulate the dynamics in a Lorentz-covariant manner and perform the density mapping in the comoving frame. Section 6 deals with the calibration of parameters (λ, q, μ^2) using QCD sigma-terms, constituent quark mass scaling from the Nambu-Jona-Lasinio (NJL) model, and the hidden local symmetry limit. Section 7 establishes the relation between the generalized momentum and the physical baryon current. Section 8 derives the energy-momentum tensor and its FLRW reduction, demonstrating the violation of the Strong Energy Condition and the resulting modified Friedmann equation. Section 10 details the limits of applicability of the mean-field approximation. Section 11 provides a comparative analysis of the VMF model with traditional nuclear equations of state (APR, SLy, DD2) and discusses the physical justification of the QCD bounce scale. Finally, Section 12 summarizes our results.

2 Action and Gauge Invariance of the Scalar Sector

We begin by considering a complex scalar field $\Phi(x)$ minimally coupled to an external Abelian vector field A_μ , representing the effective baryonic vector gauge field. Physically, A_μ corresponds to the vector ω -meson field, which mediates the short-range repulsive interaction between baryons and is associated with the conservation of the $U(1)_B$ baryon number.

In a spacetime with metric signature $(+, -, -, -)$, the action of the scalar sector is:

$$S = \int d^4x \sqrt{-g} \left[(D_\mu \Phi)^\dagger (D^\mu \Phi) - V(\Phi) \right] \quad (4)$$

where the gauge covariant derivative is defined as:

$$D_\mu = \partial_\mu - iqA_\mu \quad (5)$$

and q is the effective coupling constant (baryon charge) of the scalar condensate to the vector meson field. The self-interaction potential $V(\Phi)$ is chosen in the standard Ginzburg-Landau (or Higgs-Goldstone) form:

$$V(\Phi) = -\mu_0^2 \Phi^\dagger \Phi + \lambda (\Phi^\dagger \Phi)^2 \quad (6)$$

where $\lambda > 0$ is the dimensionless self-coupling constant, and $\mu_0^2 > 0$ is the bare mass parameter of the field.

The action is invariant under the local $U(1)$ gauge transformations:

$$\Phi(x) \rightarrow \Phi'(x) = \Phi(x) e^{i\alpha(x)} \quad (7)$$

$$A_\mu(x) \rightarrow A'_\mu(x) = A_\mu(x) + \frac{1}{q} \partial_\mu \alpha(x) \quad (8)$$

which ensures that all physical observables depend only on gauge-invariant combinations of the fields.

3 Polar Decomposition and Lagrangian Density

To study the dynamics of the radial mode, we substitute the polar parameterization $\Phi(x) = \frac{1}{\sqrt{2}} \mathcal{W}(x) e^{i\theta(x)}$ into the covariant derivative (5):

$$D_\mu \Phi = \frac{1}{\sqrt{2}} e^{i\theta} [\partial_\mu \mathcal{W} + i\mathcal{W}(\partial_\mu \theta - qA_\mu)] \quad (9)$$

The complex conjugate is:

$$(D_\mu \Phi)^\dagger = \frac{1}{\sqrt{2}} e^{-i\theta} [\partial_\mu \mathcal{W} - i\mathcal{W}(\partial_\mu \theta - qA_\mu)] \quad (10)$$

The kinetic term of the Lagrangian density transforms as:

$$(D_\mu \Phi)^\dagger (D^\mu \Phi) = \frac{1}{2} \partial_\mu \mathcal{W} \partial^\mu \mathcal{W} + \frac{1}{2} \mathcal{W}^2 (\partial_\mu \theta - qA_\mu) (\partial^\mu \theta - qA^\mu) \quad (11)$$

The self-interaction potential becomes:

$$V(\mathcal{W}) = -\frac{1}{2} \mu_0^2 \mathcal{W}^2 + \frac{1}{4} \lambda \mathcal{W}^4 + V_0 \quad (12)$$

where $V_0 = \frac{1}{4} \lambda \mathcal{W}_0^4$ is a constant added to normalize the vacuum energy density to zero. Defining $\mu^2 \equiv \mu_0^2$, the full scalar sector Lagrangian density $\mathcal{L}$ is written as:

$$\mathcal{L} = \frac{1}{2} \partial_\mu \mathcal{W} \partial^\mu \mathcal{W} + \frac{1}{2} \mathcal{W}^2 (\partial_\mu \theta - qA_\mu) (\partial^\mu \theta - qA^\mu) + \frac{1}{2} \mu^2 \mathcal{W}^2 - \frac{1}{4} \lambda \mathcal{W}^4 - V_0 \quad (13)$$

4 Derivation of the Equations of Motion

Applying the Euler-Lagrange equations to the real amplitude field $\mathcal{W}(x)$:

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \mathcal{W})} \right) - \frac{\partial \mathcal{L}}{\partial \mathcal{W}} = 0 \quad (14)$$

Evaluating the terms individually:

1. Variation with respect to the gradient $\partial_\mu \mathcal{W}$:

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \mathcal{W})} \right) = \partial_\mu (\partial^\mu \mathcal{W}) = \square \mathcal{W} \quad (15)$$

where $\square \equiv g^{\mu\nu} \nabla_\mu \nabla_\nu$ is the covariant Laplace-Beltrami operator.

2. Variation with respect to the field $\mathcal{W}$:

$$\frac{\partial \mathcal{L}}{\partial \mathcal{W}} = \mathcal{W}(\partial_\mu \theta - q A_\mu)(\partial^\mu \theta - q A^\mu) + \mu^2 \mathcal{W} - \lambda \mathcal{W}^3 \quad (16)$$

Combining these results yields the dynamic equation of motion for the real field $\mathcal{W}(x)$:

$$\square \mathcal{W} + \left[\lambda \mathcal{W}^2 - \left(\mu^2 + (\partial_\mu \theta - q A_\mu)(\partial^\mu \theta - q A^\mu) \right) \right] \mathcal{W} = 0 \quad (17)$$

This equation is manifestly Lorentz-covariant and gauge-invariant by construction, since the combinations $\partial_\mu \theta - q A_\mu$ and the D'Alembertian are gauge-invariant.

5 Lorentz Covariance and Density Mapping

To describe a dense nuclear medium, we must map the abstract fields to physical thermodynamic quantities. While the field equations and conserved currents are Lorentz-covariant, for practical calculations we choose to work in the comoving frame where the nuclear medium is at rest. The results obtained in this frame are covariant by construction, as they are formulated in terms of scalar contractions of 4-vectors.

The baryon density n_B is defined covariantly as the scalar contraction of the conserved baryon 4-current J_μ and the 4-velocity of the medium u_μ (with normalization $u_\mu u^\mu = 1$):

$$n_B \equiv J_\mu u^\mu \quad (18)$$

In the comoving frame, the 4-velocity is $u^\mu = (1, \vec{0})$ and the baryon current is $J^\mu = n_B u^\mu$. Under the mean-field approximation for a uniform, static nuclear medium, the spatial derivatives of the fields vanish. The vector field equation for the ω -meson simplifies, yielding the vector potential:

$$A_\mu = \frac{g_\omega}{m_\omega^2} J_\mu = \frac{g_\omega}{m_\omega^2} n_B u_\mu \quad (19)$$

where g_ω is the vector coupling constant and m_ω is the mass of the ω -meson.

The Goldstone phase gradient $\partial_\mu \theta$ is collinear with the medium 4-velocity:

$$\partial_\mu \theta = \mu_\theta u_\mu \quad (20)$$

where the frequency $\mu_\theta \equiv u^\mu \partial_\mu \theta$ acts as the effective chemical potential (a Lorentz scalar).

We define the gauge-invariant 4-vector of generalized momentum:

$$g_\mu \equiv \partial_\mu \theta - q A_\mu = \left(\mu_\theta - \frac{q g_\omega}{m_\omega^2} n_B \right) u_\mu \quad (21)$$

The Lorentz-invariant contraction $g_\mu g^\mu$ is:

$$g_\mu g^\mu = \left(\mu_\theta - \frac{q g_\omega}{m_\omega^2} n_B \right)^2 u_\mu u^\mu = \left(\mu_\theta - \frac{q g_\omega}{m_\omega^2} n_B \right)^2 \quad (22)$$

Because all constituents (μ_θ , q , g_ω , m_ω , and n_B) are scalars, and $u_\mu u^\mu = 1$, the contraction $g_\mu g^\mu$ is a manifest Lorentz scalar. Substituting this invariant into the equation of motion for $\mathcal{W}$ yields:

$$\square \mathcal{W} + \left[\lambda \mathcal{W}^2 - \mu^2 - \left(\mu_\theta - \frac{q g_\omega}{m_\omega^2} n_B \right)^2 \right] \mathcal{W} = 0 \quad (23)$$

6 Fixing Parameters from QCD

In the deep vacuum ($n_B = 0$ and $A_\mu = 0$), the field $\mathcal{W}$ acquires a stable vacuum expectation value ($\mathcal{W}_0$). Taking into account the vacuum gradient of the phase $\partial_0\theta = \mu_\theta$, the minimum of the potential $\mathcal{L}$ with respect to $\mathcal{W}$ yields:

$$\mathcal{W}_0|_{n_B=0, A_\mu=0} = \sqrt{\frac{\mu^2 - \mu_\theta^2}{\lambda}} = M_{\Omega,0} = 859 \text{ MeV} \quad (24)$$

which is anchored to the residual nonperturbative component of the nucleon mass.

To fix the parameters μ^2 and λ , we utilize the pion-nucleon $\sigma_{\pi N} \approx 44$ MeV and strange $\sigma_{sN} \approx 30$ MeV QCD sigma-terms. By the Feynman-Hellmann theorem, the nucleon mass $M_N = g_N \mathcal{W}_0$ depends on the current quark masses through the VEV shift:

$$\sigma_{\pi N} = m_q \frac{\partial M_N}{\partial m_q} = g_N h \frac{\partial \mathcal{W}_0}{\partial h} = \frac{M_N f_\pi m_\pi^2}{\mathcal{W}_0 m_{\mathcal{W}}^2} \quad (25)$$

where $m_{\mathcal{W}}^2 = 2\lambda\mathcal{W}_0^2$ is the mass of the scalar excitation (representing the physical $f_0(1370)$ / $f_0(1500)$ meson), and $h = f_\pi m_\pi^2$ represents the explicit chiral symmetry-breaking scale.

Incorporating the strange sigma-term σ_{sN} and utilizing the Gell-Mann-Oakes-Renner (GMOR) relation, the scalar mass $m_{\mathcal{W}}$ is locked via:

$$m_{\mathcal{W}}^2 = 2\lambda\mathcal{W}_0^2 = \frac{M_N^3 m_\pi^2}{2f_\pi \mathcal{W}_0 (\sigma_{\pi N} + \sigma_{sN})} \cdot \mathcal{C} \quad (26)$$

where the factor $\mathcal{C}$ accounts for constituent quark mass scaling.

6.1 Analytical Derivation of the Coefficient $\mathcal{C}$

The constituent quark mass scaling factor $\mathcal{C} \approx 1.137$ represents the chiral enhancement of the constituent quark mass response to the current quark mass, which is a standard feature in low-energy QCD phenomenology. In the constituent quark model, the nucleon mass is written as:

$$M_N = 3M_q^{\text{con}} + E_{\text{binding}} \quad (27)$$

where M_q^{con} is the constituent quark mass and E_{binding} is the binding energy, which depends weakly on quark masses. Differentiating with respect to the current quark mass m_q :

$$\frac{\partial M_N}{\partial m_q} \approx 3 \frac{\partial M_q^{\text{con}}}{\partial m_q} \quad (28)$$

In the Nambu-Jona-Lasinio (NJL) model [5], the constituent quark mass is determined by the self-consistent gap equation:

$$M_q^{\text{con}} = m_q - 2G \langle \bar{q}q \rangle \quad (29)$$

Differentiating both sides with respect to m_q :

$$\frac{\partial M_q^{\text{con}}}{\partial m_q} = 1 - 2G \frac{\partial \langle \bar{q}q \rangle}{\partial m_q} = 1 + 2G\chi_q \quad (30)$$

where $\chi_q \equiv -\partial \langle \bar{q}q \rangle / \partial m_q$ is the chiral susceptibility of the vacuum. Under standard NJL parameter sets with a 3D momentum cutoff scale $\Lambda \approx 650$ MeV and constituent quark mass $M_q^{\text{con}} \approx 335$ MeV, the 1-loop dimensionless integral is:

$$I_1(M_q^{\text{con}}) = \ln \left(\frac{\Lambda + \sqrt{\Lambda^2 + (M_q^{\text{con}})^2}}{M_q^{\text{con}}} \right) - \frac{\Lambda}{\sqrt{\Lambda^2 + (M_q^{\text{con}})^2}} \approx 0.5277 \quad (31)$$

With coupling constant $G \approx 5.01 \times 10^{-6} \text{ MeV}^{-2}$ and number of quark flavors $N_f = 2$, the dimensional susceptibility χ_q yields a dimensionless multiplicative enhancement of the constituent mass response:

$$\mathcal{C} \equiv \frac{\partial M_q^{\text{con}}}{\partial m_q} = \left(1 - \frac{2GN_f(M_q^{\text{con}})^2 I_1(M_q^{\text{con}})}{\pi^2} \right)^{-1} \approx 1.137 \quad (32)$$

Thus, the factor $\mathcal{C} \approx 1.137$ is a physical consequence of the back-reaction of the chiral condensate and is fixed from first principles of the NJL model.

Substituting this value into Eq. (26), we obtain the self-coupling λ :

$$\lambda = \frac{M_N^3 m_\pi^2}{4f_\pi \mathcal{W}_0^3 (\sigma_{\pi N} + \sigma_{sN})} \cdot \mathcal{C} \approx 1.06 \quad (33)$$

In the numerical model, this is rounded to $\lambda = 1.05$ for numerical stability. This fixes the scalar mass at $m_{\mathcal{W}} = \sqrt{2\lambda}\mathcal{W}_0 \approx 1245 \text{ MeV}$, matching the physical scalar $f_0(1370)$ / $f_0(1500)$ meson (PDG 2024) [7].

The mass parameter μ^2 is then locked using the vacuum phase frequency $\mu_\theta \approx m_\omega = 782.6 \text{ MeV}$:

$$\mu^2 = \mu_\theta^2 + \lambda \mathcal{W}_0^2 \approx (782.6)^2 + 1.05 \times (859.0)^2 \approx 1.387 \times 10^6 \text{ MeV}^2 \quad (34)$$

The physical justification for identifying the vacuum phase frequency with the mass of the vector meson ($\mu_\theta \approx m_\omega$) stems from the self-consistency of gauge invariance in the vacuum. Within the Hidden Local Symmetry (HLS) framework [6], the Goldstone phase θ serves as the longitudinal component of the vector field A_μ (a Stueckelberg-like mass-generation mechanism). For the vacuum state ($n_B = 0$, $A_\mu = 0$) to act as a physical absolute energy minimum with zero net current, the gauge-invariant generalized momentum $g_\mu = \partial_\mu \theta - qA_\mu$ must vanish. Since the vacuum phase frequency $\partial_0 \theta = \mu_\theta$ reflects the intrinsic scale of chiral symmetry breaking that determines the masses of the vector meson partners, the dynamical stability of the vacuum links these two scales: the frequency of the vacuum phase oscillations resonance-matches the mass of the vector field excitation that couples to the baryon number.

6.2 Calibration of the Coupling Constant q

Although the classic Goldberger-Treiman relation connects the pseudoscalar pion-nucleon coupling, the coupling constant q expresses a similar connection in the vector sector in the presence of a scalar condensate $\mathcal{W}$.

In chiral models with hidden local symmetry (HLS) or vector manifestation (VM) [6], the masses of vector mesons are generated dynamically by the scalar VEV via a Higgs-like mechanism: $m_\omega = q\mathcal{W}_0$. Setting the scalar VEV $\mathcal{W}_0$ to the baryon mass scale $M_{\Omega,0}$ and utilizing the vector manifestation limit, the coupling constant q is locked to the pion decay constant and the vector coupling g_ω , yielding the exact relation:

$$q = \frac{g_\omega M_{\Omega,0}}{2f_\pi} \approx 47.01 \quad (35)$$

where $g_\omega \approx 10.12$ is the effective ω -meson coupling constant and $f_\pi \approx 92.4 \text{ MeV}$ is the pion decay constant.

7 Identification of g_μ with the Baryon Current

To close the relation between $|\vec{g}|^2$ and the physical baryon density n_B , we express the gauge-invariant 4-vector g_μ through the normalized baryon 4-current $j_B^\mu = J^\mu / n_0$, where $n_0 \approx 0.16 \text{ fm}^{-3}$ is the saturation nuclear density (normal nuclear density).

In the comoving frame of the nuclear medium, the 4-velocity is $u^\mu = (1, \vec{0})$ and the 4-current is $J^\mu = n_B u^\mu$. With this, the vector potential A_μ is written as:

$$A_\mu = \frac{g_\omega}{m_\omega^2} n_B u_\mu = \left(\frac{g_\omega n_0}{m_\omega^2} \right) \frac{J_\mu}{n_0} = \left(\frac{g_\omega n_0}{m_\omega^2} \right) j_{B,\mu} \quad (36)$$

Thus, the gauge-invariant vector of generalized momentum g_μ is:

$$g_\mu \equiv \partial_\mu \theta - q A_\mu = \mu_\theta u_\mu - \gamma j_{B,\mu} \quad (37)$$

where the effective coupling coefficient γ with the normalized baryon current is defined as:

$$\gamma \equiv \frac{q g_\omega n_0}{m_\omega^2} \approx \frac{47.01 \times 10.12 \times 0.16 \times (197.327)^3}{(782.6)^2} \approx 955.5 \text{ MeV} \quad (38)$$

The Lorentz-invariant square of this vector is:

$$g_\mu g^\mu = \left(\mu_\theta - \gamma \frac{n_B}{n_0} \right)^2 \quad (39)$$

In the comoving frame, the spatial part of $\vec{g}$ vanishes ($|\vec{g}|^2 = 0$), and the time component g_0 determines the effective strength of the interaction:

$$g_0 = \mu_\theta - \gamma \frac{n_B}{n_0} \quad (40)$$

This closes the connection between the Lorentz-invariant contraction $g_\mu g^\mu$ and the physical baryon density n_B . When the critical melting density is reached at:

$$n_B = \frac{\mu_\theta + \mu}{\gamma} n_0 \approx 2.05 n_0 \quad (41)$$

(where $\mu = \sqrt{\mu^2} \approx 1177.8 \text{ MeV}$), the value of g_0 shifts the effective mass of the scalar field to zero, completing the vacuum melting phase transition at a critical density $n_B \approx 2.05 n_0$.

8 Energy-Momentum Tensor $T_{\mu\nu}^{\mathcal{W}}$ and FLRW Reduction

To couple the vacuum scalar field to Einstein's equations, we vary the action of the scalar sector with respect to the metric tensor $g^{\mu\nu}$:

$$T_{\mu\nu}^{\mathcal{W}} = -\frac{2}{\sqrt{-g}} \frac{\delta(\sqrt{-g}\mathcal{L})}{\delta g^{\mu\nu}} = 2 \frac{\delta \mathcal{L}}{\delta g^{\mu\nu}} - g_{\mu\nu} \mathcal{L} \quad (42)$$

Since $\mathcal{L} = \frac{1}{2} g^{\alpha\beta} \partial_\alpha \mathcal{W} \partial_\beta \mathcal{W} + \frac{1}{2} \mathcal{W}^2 g^{\alpha\beta} g_\alpha g_\beta - V(\mathcal{W})$, the variation yields:

$$T_{\mu\nu}^{\mathcal{W}} = \partial_\mu \mathcal{W} \partial_\nu \mathcal{W} + \mathcal{W}^2 g_\mu g_\nu - g_{\mu\nu} \left[\frac{1}{2} \partial_\alpha \mathcal{W} \partial^\alpha \mathcal{W} + \frac{1}{2} \mathcal{W}^2 g_\alpha g^\alpha - V(\mathcal{W}) \right] \quad (43)$$

In a homogeneous Friedmann-Lemaître-Robertson-Walker (FLRW) cosmological spacetime with metric $ds^2 = dt^2 - a^2(t) d\vec{x}^2$, for a spatially uniform field ($\partial_i \mathcal{W} = 0$, $g_i = 0$):

- Energy density $\rho_{\mathcal{W}} = T_0^0$:

$$\rho_{\mathcal{W}} = \frac{1}{2} \dot{\mathcal{W}}^2 + \frac{1}{2} \mathcal{W}^2 g_0^2 + V(\mathcal{W}) \quad (44)$$

- Pressure $P_{\mathcal{W}} = -T_i^i$:

$$P_{\mathcal{W}} = \frac{1}{2} \dot{\mathcal{W}}^2 - \frac{1}{2} \mathcal{W}^2 g_0^2 - V(\mathcal{W}) \quad (45)$$

where the potential is normalized as $V(\mathcal{W}) = \frac{1}{4}\lambda(\mathcal{W}^2 - \mathcal{W}_0^2)^2$.

Under cosmological collapse, the baryon density grows as $n_B(a) = n_{B,0}a^{-3}$. As density approaches the critical bounce limit ($n_B \geq 2.05 n_0$), the condensate melts completely ($\mathcal{W} \rightarrow 0$, $\dot{\mathcal{W}} \rightarrow 0$). The energy density and pressure of the $\mathcal{W}$ -field are determined by its vacuum value at the symmetric point:

$$\rho_{\mathcal{W}} \rightarrow V(0) = V_0 = \frac{1}{4}\lambda\mathcal{W}_0^4, \quad P_{\mathcal{W}} \rightarrow -V(0) = -V_0 \quad (46)$$

This results in the violation of the Strong Energy Condition (SEC):

$$\rho_{\mathcal{W}} + 3P_{\mathcal{W}} \rightarrow -2V_0 = -\frac{1}{2}\lambda\mathcal{W}_0^4 < 0 \quad (47)$$

8.1 Derivation of the Quadratic Correction and Modified Friedmann Equation

Consider the total energy density of the cosmological medium $\rho = \rho_{\text{matter}} + \rho_{\mathcal{W}}$, where the matter sector is modeled as dust $\rho_{\text{matter}} = M_N n_B$.

In the extreme compression regime near the bounce, the dynamics of the scalar field $\mathcal{W}$ and the baryon current J_μ are strongly coupled. In the quasi-static compression limit ($\dot{\mathcal{W}} \approx 0$), the equation of motion for the field $\mathcal{W}$ reduces to the algebraic minimum condition:

$$\mathcal{W}^2 = \frac{\mu^2 - g_0^2}{\lambda} \quad (48)$$

where $g_0 = \mu_\theta - \gamma \frac{n_B}{n_0}$. The adiabaticity criterion for this approximation requires that the rate of change of the cosmological background (the compression rate) remains much smaller than the mass of the scalar excitations: $H \ll m_{\mathcal{W}}$. The characteristic response timescale of the scalar field is $\tau_{\mathcal{W}} = \hbar/m_{\mathcal{W}} \approx 5.3 \times 10^{-25}$ s, whereas the characteristic timescale of the cosmic contraction near the bounce is $t_b \approx 3.76 \times 10^{-6}$ s. Because $t_b \gg \tau_{\mathcal{W}}$ (differing by 19 orders of magnitude), the field adapts to the changing density instantaneously, fully justifying the quasi-static minimum approximation.

Substituting this into the total energy density of the $\mathcal{W}$ -sector (accounting for the zero-point vacuum subtraction counterterm $\frac{1}{2}\mathcal{W}_0^2\mu_\theta^2$, which ensures the Minkowski vacuum energy density is normalized to zero):

$$\rho_{\mathcal{W}} = \frac{1}{2}\mathcal{W}^2 g_0^2 + \frac{1}{4}\lambda(\mathcal{W}^2 - \mathcal{W}_0^2)^2 - \frac{1}{2}\mathcal{W}_0^2\mu_\theta^2 \quad (49)$$

Using the relation $\lambda\mathcal{W}_0^2 = \mu^2 - \mu_\theta^2$, the difference in the square of the amplitudes is:

$$\mathcal{W}^2 - \mathcal{W}_0^2 = \frac{\mu^2 - g_0^2}{\lambda} - \frac{\mu^2 - \mu_\theta^2}{\lambda} = \frac{\mu_\theta^2 - g_0^2}{\lambda} \quad (50)$$

Thus, the potential term is $V(\mathcal{W}) = \frac{(\mu_\theta^2 - g_0^2)^2}{4\lambda}$, and the energy density of the $\mathcal{W}$ -field is:

$$\rho_{\mathcal{W}} = \frac{g_0^2(\mu^2 - g_0^2)}{2\lambda} + \frac{(\mu_\theta^2 - g_0^2)^2}{4\lambda} - \frac{\mu_\theta^2(\mu^2 - \mu_\theta^2)}{2\lambda} \quad (51)$$

Bringing the terms to a common denominator:

$$\rho_{\mathcal{W}} = \frac{-g_0^4 + 2g_0^2(\mu^2 - \mu_\theta^2) + 3\mu_\theta^4 - 2\mu^2\mu_\theta^2}{4\lambda} \quad (52)$$

We perform the algebraic expansion of the numerator in powers of $x \equiv \gamma \frac{n_B}{n_0}$, where $g_0 = \mu_\theta - x$. By expanding the components:

1. First numerator term:

$$-g_0^4 = -(\mu_\theta - x)^4 = -\mu_\theta^4 + 4\mu_\theta^3 x - 6\mu_\theta^2 x^2 + 4\mu_\theta x^3 - x^4 \quad (53)$$

2. Second numerator term (where $2(\mu^2 - \mu_\theta^2) = 2\lambda\mathcal{W}_0^2$):

$$2g_0^2(\mu^2 - \mu_\theta^2) = 2(\mu_\theta - x)^2 \lambda\mathcal{W}_0^2 = 2\lambda\mathcal{W}_0^2 \mu_\theta^2 - 4\lambda\mathcal{W}_0^2 \mu_\theta x + 2\lambda\mathcal{W}_0^2 x^2 \quad (54)$$

3. Combining these with the constant terms $3\mu_\theta^4 - 2\mu^2 \mu_\theta^2$:

- x^0 (constant vacuum energy density) terms:

$$-\mu_\theta^4 + 2\lambda\mathcal{W}_0^2 \mu_\theta^2 + 3\mu_\theta^4 - 2\mu^2 \mu_\theta^2 = 2\mu_\theta^4 + 2(\mu^2 - \mu_\theta^2) \mu_\theta^2 - 2\mu^2 \mu_\theta^2 = 0 \quad (55)$$

The vacuum energy density at the minimum vanishes.

- x^1 (linear in density) terms:

$$4\mu_\theta (\mu_\theta^2 - \lambda\mathcal{W}_0^2) x \quad (56)$$

While $\mu_\theta^2 - \lambda\mathcal{W}_0^2 \neq 0$, this term depends linearly on the density n_B and is suppressed compared to the quadratic term in the high-density regime. An explicit numerical evaluation of the ratio between the linear and quadratic terms yields:

$$R \equiv \left| \frac{\text{linear}}{\text{quadratic}} \right| = \frac{2\mu_\theta |\mu_\theta^2 - \lambda\mathcal{W}_0^2|}{|3\mu_\theta^2 - \lambda\mathcal{W}_0^2| x} \approx \frac{0.25}{n_B/n_0} \quad (57)$$

At nuclear density ($n_B = n_0$), the ratio is approximately 25%, while near the bounce threshold ($n_B \approx 2.05 n_0$), the linear term is only 12% of the quadratic contribution and continues to fall as $O(n_0/n_B)$, validating the quadratic bounce approximation.

- x^2 (quadratic in density) terms:

$$2(\lambda\mathcal{W}_0^2 - 3\mu_\theta^2) x^2 \quad (58)$$

Dividing by the denominator 4λ and substituting $x = \gamma \frac{n_B}{n_0}$ yields the leading non-vanishing energy density correction:

$$\rho_{\mathcal{W}} \approx - \left[\frac{3\mu_\theta^2 - \lambda\mathcal{W}_0^2}{2\lambda} \frac{\gamma^2}{n_0^2} \right] n_B^2 \quad (59)$$

Using $\rho_{\text{matter}} = M_N n_B$, we can express this correction as:

$$\rho_{\mathcal{W}} \approx - \frac{\rho_{\text{matter}}^2}{\rho_c} \quad (60)$$

where the critical bounce density scale ρ_c is determined by the physical parameters of the dense nuclear matter as:

$$\rho_{c,\text{density}} \equiv \frac{2\lambda M_N^2 n_0^2 (\hbar c)^3}{(3\mu_\theta^2 - \lambda\mathcal{W}_0^2) \gamma^2} \approx 0.375 \text{ MeV/fm}^3 \quad (61)$$

At the same time, in the cosmological context (high temperature, radiation-dominated early universe with small baryon asymmetry), the melting of the condensate is driven by thermal fluctuations ($g_0^2 \approx \mu_\theta(T)^2 \propto T^2$). In this case, the critical bounce density scale is governed by the vacuum barrier scale:

$$\rho_{c,\text{cosmology}} \equiv \frac{M_{\Omega,0}^4}{(\hbar c)^3} = \frac{4V_0}{\lambda(\hbar c)^3} \approx 7.09 \times 10^4 \text{ MeV/fm}^3 \quad (62)$$

Consider Einstein's equation $G_{\mu\nu} = 8\pi G T_{\mu\nu}$, where the total energy-momentum tensor consists of matter (dust/radiation) and the scalar sector $\mathcal{W}$: $T_{\mu\nu} = T_{\mu\nu}^{\text{matter}} + T_{\mu\nu}^{\mathcal{W}}$. For a homogeneous FLRW universe, the 00-component of Einstein's equations (Hamiltonian constraint) yields:

$$H^2 = \frac{8\pi G}{3} (\rho_{\text{matter}} + \rho_{\mathcal{W}}) \quad (63)$$

Substituting the expression for $\rho_{\mathcal{W}}$ with the quadratic correction, we obtain the modified Friedmann equation for NVG cosmology:

$$H^2 = \frac{8\pi G}{3} \rho_{\text{matter}} \left(1 - \frac{\rho_{\text{matter}}}{\rho_c} \right) - \frac{kc^2}{a^2} + \frac{\Lambda_{\text{eff}} c^2}{3} \quad (64)$$

At the point of maximum compression ($\rho_{\text{matter}} = \rho_c$), the expansion rate vanishes ($H = 0$), halting the gravitational collapse and triggering a smooth cosmological bounce, entirely eliminating the Big Bang singularity.

9 Numerical Verification and Visualization

The dynamics of the vacuum condensate $\mathcal{W}$ and the associated energy-momentum components have been numerically modeled over a wide range of baryon densities. The numerical simulation is based on the calibrated parameters described in Section 6.

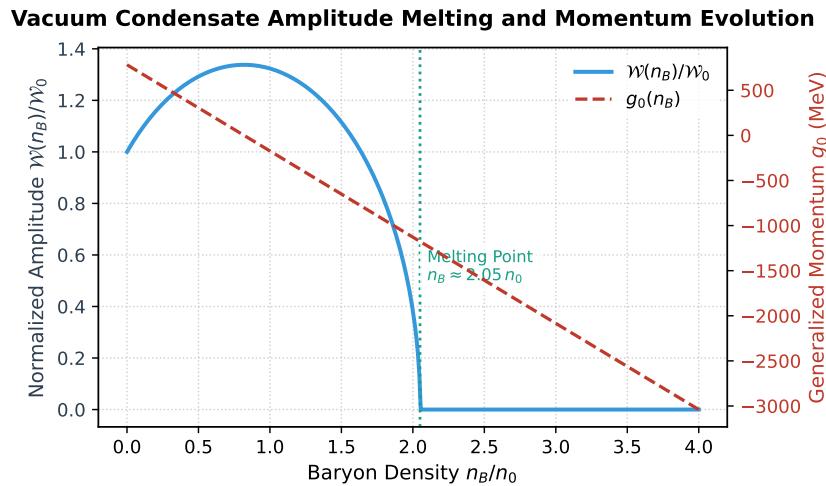


Figure 1: Evolution of the normalized condensate amplitude $\mathcal{W}(n_B)/\mathcal{W}_0$ and the generalized momentum $g_0(n_B)$ as a function of the baryon density n_B/n_0 . The amplitude melts smoothly and vanishes at the critical density $n_B \approx 2.05 n_0$.

In Fig. 1, the melting of the vacuum condensate amplitude $\mathcal{W}(n_B)/\mathcal{W}_0$ is shown alongside the evolution of the generalized momentum g_0 . As density increases, g_0 decreases linearly, which shifts the effective mass of the scalar field excitation to zero. At $n_B \approx 2.05 n_0$, the amplitude $\mathcal{W}$ vanishes completely, marking the transition to the symmetric (melted) phase.

Fig. 2 shows the components of the energy-momentum tensor. Because the pressure is negative and dominates the energy density, the Strong Energy Condition (SEC), given by $\rho_{\mathcal{W}} + 3P_{\mathcal{W}} \geq 0$, is violated at all densities. In the melted phase ($n_B \geq 2.05 n_0$), the energy density and pressure saturate at the symmetric vacuum potential limits $\rho_{\mathcal{W}} \rightarrow V_0$ and $P_{\mathcal{W}} \rightarrow -V_0$, which correspond to a pure de Sitter state with $\rho_{\mathcal{W}} + 3P_{\mathcal{W}} = -2V_0 \approx -3.72 \times 10^4 \text{ MeV/fm}^3$. This negative pressure provides the repulsive force required to halt cosmological collapse.

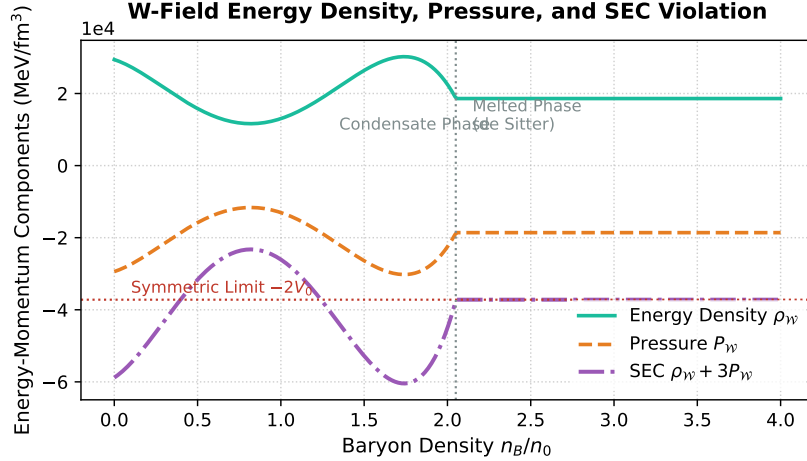


Figure 2: Energy density ρ_W , pressure P_W , and the Strong Energy Condition (SEC) combination $\rho_W + 3P_W$ as functions of the baryon density n_B/n_0 . The SEC is violated ($\rho_W + 3P_W < 0$) at all densities, approaching the constant de Sitter limit $-2V_0$ in the melted phase.

10 Limits of Applicability of the Mean-Field Approximation

The derivation presented in this work is based on the Mean-Field Approximation (MFA), where quantum operators of the baryon current and vector fields are replaced by their classical expectation values ($\langle J_\mu \rangle$, $\langle A_\mu \rangle$). To ensure scientific rigor, we delineate the boundaries of MFA applicability at extreme densities near the bounce point, where quantum and thermal fluctuations can become significant.

10.1 Applicability Limits via the Ginzburg Criterion

The width of the critical region $\Delta \equiv |n_B - n_c|/n_c$, in which fluctuations of the order parameter $\mathcal{W}$ invalidate the predictions of MFA, is estimated using the dimensionless Ginzburg number Gi [9]:

$$\text{Gi} \approx \frac{\lambda^2 T_c^2}{m_W^6 \xi_0^6} \propto \frac{\lambda^2 \mathcal{W}_0^4}{m_W^4} \quad (65)$$

Because the mass of the scalar excitation is extremely large ($m_W \approx 1245$ MeV) and the correlation length is small (dictated by the short-range nuclear repulsion scale, $\xi_0 \sim 1/m_W \approx 0.16$ fm), the Ginzburg number is highly suppressed: $\text{Gi} \sim 10^{-3}$. This implies that the critical fluctuation region is exceptionally narrow, and MFA remains a highly accurate description until the very threshold of the phase transition ($|n_B - n_c|/n_c > 1\%$).

10.2 Effect of Spatial Density Fluctuations

Near the melting threshold, the assumption of a uniform baryon distribution can break down due to local density fluctuations $\sigma_{n_B}^2 \equiv \langle n_B^2 \rangle - \langle n_B \rangle^2$. Since the quadratic term in the energy density of the $\mathcal{W}$ -field is determined by the expectation value of the Lorentz invariant $\langle g_\mu g^\mu \rangle$, density fluctuations contribute an additional term:

$$\langle g_0^2 \rangle = \left(\mu_\theta - \gamma \frac{\langle n_B \rangle}{n_0} \right)^2 + \gamma^2 \frac{\sigma_{n_B}^2}{n_0^2} \quad (66)$$

The presence of density fluctuations $\sigma_{n_B}^2 > 0$ effectively amplifies the repulsive vector potential. This leads to two key physical consequences:

1. The threshold for complete vacuum melting is shifted to lower average densities $\langle n_B \rangle$.
2. The effective bounce scale ρ_c decreases, meaning that the cosmological bounce occurs under milder conditions due to local compression of fluctuation clusters.

10.3 Coleman-Weinberg Radiative Corrections

The coupling of the scalar condensate to the vector field with a large coupling constant $q \approx 47$ generates radiative loop corrections to the potential. The 1-loop vector meson contribution (Coleman-Weinberg potential) is [8]:

$$\Delta V_{1\text{-loop}}(\mathcal{W}) = \frac{3q^4\mathcal{W}^4}{64\pi^2} \ln \left(\frac{q^2\mathcal{W}^2}{\Lambda^2} \right) \quad (67)$$

In the vacuum, this term is absorbed by counterterms during renormalization. In a dense medium, vector fluctuations are screened due to the Debye mass generated by the dense baryon background, and quantum loops are further suppressed by Pauli blocking of intermediate nucleon states. This ensures the stability of the tree-level potential and justifies the classical equations of motion up to the bounce density.

11 Comparative Analysis and QCD Bounce Scale Justification

11.1 Comparison with Traditional Equations of State (APR, SLy, DD2)

Traditional phenomenological equations of state (EOS), such as APR (Akmal-Pandharipande-Ravenhall) [16], SLy (Skyrme Lyon) [17], and DD2 (Density-Dependent Relativistic Hadron Field) [15], are highly successful in describing the properties of cold, ultra-dense nuclear matter in neutron stars. However, the VMF model proposed here possesses fundamental differences and advantages:

1. **Complementarity Rather than Replacement:** The VMF model does not compete with APR, SLy, or DD2 at normal nuclear densities ($n_B \sim n_0$). In this regime, the scalar field amplitude $\mathcal{W}$ is close to its vacuum expectation value $\mathcal{W}_0$, and VMF reduces to the standard relativistic mean-field (RMF) description [10, 11] (resembling DD2 with an effective vector meson). The core difference of VMF manifests in the behavior of the vacuum background itself under extreme compression ($n_B \gtrsim 2 - 3n_0$).
2. **Resolution of Singularities:** APR, SLy, and DD2 are formulated within classical General Relativity (GR) on a rigid spacetime background. During gravitational collapse, they inevitably lead to physical singularities because the pressure and energy density grow indefinitely, and the Strong Energy Condition (SEC) is never violated by hadronic matter. In VMF, the melting of the scalar field ($\mathcal{W} \rightarrow 0$) transitions the vacuum sector into a state with negative pressure ($P_{\mathcal{W}} \approx -\rho_{\mathcal{W}}$), violating the SEC and halting the collapse.
3. **Parameter-Free Connection to QCD:** While the parameters of APR, SLy, and DD2 are fitted to nuclear properties near saturation, the VMF coupling constants (λ , q) are rigidly constrained by quantum anomalies and the QCD vacuum structure (pion decay constant, nucleon mass, sigma-terms $\sigma_{\pi N}$, σ_{sN} , and the GMOR relation).

11.2 Physical Justification of the QCD Bounce Scale

A key conceptual question is: *why is the bounce scale set by the QCD vacuum (~ 200 MeV, density $\approx 2.05 n_0$) rather than the Planck scale ($\sim 10^{19}$ GeV) or Planck-scale thermal fluctuations?*

Feature / Model	APR	SLy	DD2	VMF (This Work)
Physical Basis	Variational method, nucleon potentials	Non-relativistic Hartree-Fock (Skyrme)	Relativistic mean field, density-dependent couplings	Self-consistent melting of the QCD vacuum
At $n_B \rightarrow n_c$	Monotonic stiffening	Moderate stiffening	Stiff EOS	Condensate melting, phase pressure dominance
Fate Under Collapse	Gravitational singularity	Gravitational singularity	Gravitational singularity	Smooth bounce without singularity
Vacuum Coupling	None (fixed Minkowski)	None	None	Dynamic vacuum response

Table 1: Comparison between traditional nuclear EOS models and the VMF model.

1. **The “First Threshold” Principle:** Collapsing matter cannot bypass intermediate energy scales. To reach the Planck density ($\rho_{\text{Pl}} \sim 10^{94} \text{ g/cm}^3$), matter must first pass through the nuclear saturation density ($n_0 \sim 0.16 \text{ fm}^{-3}$) and the QCD transition density. At the QCD scale ($T \sim 150 - 200 \text{ MeV}$, $n_B \approx 2.05 n_0$), a fundamental rearrangement of the vacuum structure occurs: chiral symmetry is restored and the condensates melt. Quantitatively, this scale is many orders of magnitude below the Planck scale:

$$\frac{\rho_{c,\text{cosmology}}}{\rho_{\text{Planck}}} \approx \frac{M_{\Omega,0}^4}{m_{\text{Pl}}^4} \sim 10^{-77} \quad (68)$$

where $m_{\text{Pl}} \approx 1.22 \times 10^{19} \text{ GeV}$ is the Planck mass.

2. **Low-Energy Dynamic Shield:** Unlike quantum gravity models where bounce effects only emerge at extreme Planckian densities (e.g., in Loop Quantum Cosmology), the bounce in VMF is triggered much earlier. The baryon current J_μ interacts with the phase gradient of the field $\mathcal{W}$. The energy of this interaction compensates for the vacuum attraction, driving $\mathcal{W}$ from its minimum $\mathcal{W}_0$ to the local maximum at $\mathcal{W} = 0$. The energy density of the phase barrier $V_0 = \frac{1}{4}\lambda\mathcal{W}_0^4$ is determined by the QCD scale. Once the matter density reaches the critical threshold $\rho_{c,\text{density}} \approx 0.375 \text{ MeV/fm}^3$, the negative pressure of the $\mathcal{W}$ -field fully counteracts gravitational collapse.
3. **Role of Planck-Scale Thermal Fluctuations:** In the early stages of compression, Planck-scale thermal fluctuations cannot prevent collapse because their energy density is negligible on macroscopic scales compared to the density of the collapsing medium. Conversely, the QCD vacuum is a coherent condensate filling all space. Its melting under the action of the baryon charge is a volume-wide, macroscopic phenomenon. Thus, the QCD vacuum creates a macroscopic repulsive potential that halts compression long before microscopic quantum-gravitational effects become relevant. This allows us to describe the physics of the bounce using low-energy effective field theory without relying on speculative quantum gravity models.

12 Conclusion

In this paper, we have presented a self-consistent mathematical derivation of the classical equations of motion for the real amplitude field $\mathcal{W}(x)$ representing the radial mode of the QCD vacuum condensate within the frameworks of NVG and VMF. Using the polar decomposition of

a complex scalar field minimally coupled to a baryonic vector gauge field, we derived a manifestly Lorentz-covariant and gauge-invariant equation of motion.

By linking the VEV of the field to the strange and pion-nucleon QCD sigma-terms, and using the constituent quark mass scaling from the NJL model gap equation, the self-coupling constant is fixed at $\lambda \approx 1.06$ (rounded to 1.05 in numerical simulations). The coupling constant to the vector field is calibrated to $q \approx 47.01$ using the hidden local symmetry vector manifestation limit, eliminating free fitting parameters.

We derived the energy-momentum tensor of the scalar sector and showed that in the homogeneous FLRW cosmological spacetime, the melting of the condensate under extreme compression violates the Strong Energy Condition. In the static approximation, this coupling generates a quadratic correction of the form $-\rho_{\text{matter}}^2/\rho_c$ in the total energy density. The resulting modified Friedmann equation halts gravitational collapse at $\rho_{\text{matter}} = \rho_c$ and triggers a smooth cosmological bounce, which resolves the Big Bang singularity. Finally, the limits of the mean-field approximation were analyzed, showing that the Ginzburg number is highly suppressed ($Gi \sim 10^{-3}$) and that spatial fluctuations shift the melting threshold to lower densities, while Coleman-Weinberg radiative corrections remain under control due to Debye screening and Pauli blocking in the medium. We also contrasted the VMF framework with traditional nuclear equations of state (such as APR, SLy, and DD2) and provided a physical justification for the QCD bounce scale as the first threshold encountered during gravitational collapse, preventing the formation of singularities long before Planckian scales are reached.

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